Woojung Song

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Education

Seoul National University, Seoul, South Korea — Sep 2025 – Present
Integrated M.S.–Ph.D., Graduate School of Data Science
Advisor: Yohan Jo, HOLI Lab (Human-Oriented Language Intelligence)

Hanyang University, Seoul, South Korea — Mar 2022 – Aug 2025
B.I.S. Department of Information Systems — Overall GPA: 4.35/4.5 | Major GPA: 4.35/4.5

Research Interests

Understanding and shaping agent behavior to better align it with human needs and values.

- **Agent Behavior:** Behavioral evaluation, decision-making, tool use, user interaction, and value alignment.
- **Human-like Agents:** Modeling the values, personalities, and changing behavior of people and fictional characters across situations and over time.

Publications

*Equal contribution. **First** denotes first authorship, including equal contribution.

Preprints

Source Preference in the Wild: How LLM Agents Favor Items by Source, and How to Reduce It
Jonghyun Song*, Haewon Park*, Jeonghoon Shim*, **Woojung Song**, Yohan Jo
Under Review (2026)

AgentHabit: Characterizing Distinct Behaviors of Agents on Everyday Tasks
Woojung Song*, Hoyeol Yang*, Jeonghoon Shim, Sungjib Lim, Jonggeun Lee, Yunho Choi, Yohan Jo
Under Review (2026) · **First**

Agents' Overreliance on Unreliable Tools
Hoyeol Yang*, **Woojung Song***, Taewon Kim, Jonghyun Song, Seoyeon Park, Yohan Jo
Under Review (2026) · **First**

Conferences & Journals

ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time?
Woojung Song*, Nalim Kim*, Sangjun Song, Chaewon Heo, Jongwon Lim, Yohan Jo
EMNLP 2026 (Main) · **First**

Human Psychometric Questionnaires Mischaracterize LLM Behavior
Woojung Song*, Dongmin Choi*, Yoonah Park, Jongwook Han, Eun-Ju Lee, Yohan Jo
EMNLP 2026 (Main) · **First**

Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models
Jonggeun Lee*, **Woojung Song***, Jongwook Han, Haesung Pyun, Yohan Jo
ACL 2026 (Main) · **First**

Non-Collaborative User Simulators for Tool Agents
Jeonghoon Shim, **Woojung Song**, Cheyon Jin, Seungwon Kook, Yohan Jo
ICLR 2026

Psychometric Item Validation Using Virtual Respondents with Trait-Response Mediators
Sungjib Lim, **Woojung Song**, Eun-Ju Lee, Yohan Jo
TACL 2026

Quantifying Data Contamination in Psychometric Evaluations of LLMs
Jongwook Han*, **Woojung Song***, Jonggeun Lee*, Yohan Jo
EACL 2026 (Findings) · **First**

Publications (continued)

Value Portrait: Assessing Language Models' Values through Psychometrically and Ecologically Valid Items

Jongwook Han*, Dongmin Choi*, **Woojung Song***, Eun-Ju Lee, Yohan Jo

ACL 2025 (Main) · **First**

Interpretable Prediction of Private Brand Purchases by Pet Type in E-Commerce for Consumer Behavior Analysis Using Real-World Transaction Data

Jaehyuk Lee*, **Woojung Song***, Jina Kim, Yoona Chung, Eunchan Kim

PeerJ Computer Science 2026 · **First**

Experience

Seoul National University, HOLI Lab — Seoul, South Korea
Undergraduate Research Intern — Jul 2024 – Aug 2025

- Researched value alignment in Large Language Models (LLMs)
- Developed simulation frameworks using LLMs to model complex human behaviors and interactions

LG Electronics, H&A Sound/Image Intelligence R&D Lab — Seoul, South Korea
Intern — Mar 2024 – Jun 2024

- Enhanced LLM-based Web Question Answering using RAG and multi-agent approaches
- Developed WAAG, an automated QA evaluation system with 40% accuracy improvement
- Achieved highest level in LG SW Competency Certification Test

US 8th Army, Korean Augmentation To the United States Army — Pyeongtaek, South Korea
Sergeant (Senior KATUSA) — May 2020 – Nov 2021

- Served in US 8th Army as part of the mandatory military service program
- Received Certificate of Appreciation for exceptional service

Patents

English Problem Creation System Using Multi-Agents — App. No: 10-2024-0109541
Filed: August 16, 2024
AI system for generating high-quality English exam questions and reducing educator workload.

Scholarships

Presidential Science Scholarship for Graduate Students — South Korea
Ministry of Science and ICT — 2026 – 2027

- Selected at a 25:1 competition ratio

Awards

Busan Mayor's Award — Busan, South Korea
DIVE Competition, 2nd Place — 2026

- Also won 1st Place from Busan Metropolitan Corporation in the same competition

Chuncheon Mayor's Award — Chuncheon, South Korea
Chuncheon Generative AI Competition, 3rd Place — Jun 2025

- Developed "Blue-Ming" Chuncheon City Youth Entrepreneurship AI Consulting Platform

Busan Mayor's Award — Busan, South Korea
DIVE Competition, App and Service Development, 3rd Place — 2024

- Developed "Korail Donghae" application integrating Language Models with custom tools and RAG
- Also won 1st Place from Korea Railroad Corporation Busan-Gyeongnam Headquarters

Skills & Competencies

Programming: Python, C/C++ **Languages:** Korean (native), English (fluent)

Extra-curricular Activities

FORIF (Hanyang Programming Club) — Seoul, South Korea
Member (2022 – 2025), Vice President (2023)

- Organized club events, including HPEC, doubling club membership
- Managed HPEC & HSPC algorithm competition as Project Manager